

# Mini Lesson Sample Script

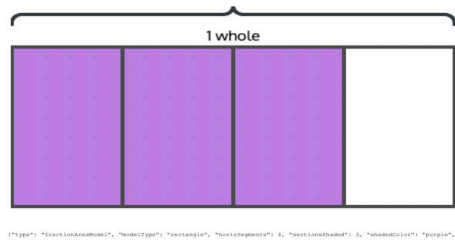
## Materials

(S) Personal white board, Mini-Lesson Worksheet

### PROBLEM

**1 Use an area model to decompose fourths into eighths, and write addition and multiplication sentences to show the equivalence.**

The rectangle below represents 1 whole and is shaded to show  $\frac{3}{4}$ .



Draw a horizontal line to decompose the fourths into eighths. Then, write an addition sentence using parentheses and a multiplication sentence to show that  $\frac{3}{4} = \frac{6}{8}$ .

*Answer:  $(\frac{1}{8} + \frac{1}{8}) + (\frac{1}{8} + \frac{1}{8}) + (\frac{1}{8} + \frac{1}{8}) = \frac{6}{8}$ ;  $6 \times \frac{1}{8} = \frac{6}{8}$ ;  $\frac{3}{4} = \frac{6}{8}$*

### SAMPLE SCRIPT

**T:** Look at the rectangle on your worksheet. It represents 1 whole. How many equal parts is it divided into?

**S:** 4 parts.

**T:** And how many are shaded?

**S:** 3 of them. So  $\frac{3}{4}$  is shaded.

**T:** Right. Now, I want to decompose each fourth into 2 equal parts to make eighths. I'm going to draw one horizontal line straight across the middle of the rectangle. Go ahead and do that on yours.

**S:** (Draw.)

**T:** How many equal parts are in the whole rectangle now?

**S:** 8 parts.

**T:** So each part is what size?

**S:**  $\frac{1}{8}$ .

**T:** How many eighths are shaded?

**S:** 6 eighths.

**T:** So  $\frac{3}{4}$  is the same as  $\frac{6}{8}$ . The shaded area didn't change — we just have more pieces, and each piece is smaller. Now let's write an addition sentence. Each fourth became 2 eighths, so we can group them with parentheses:  $(\frac{1}{8} + \frac{1}{8})$  for the first fourth, plus  $(\frac{1}{8} + \frac{1}{8})$  for the second, plus  $(\frac{1}{8} + \frac{1}{8})$  for the third. Write that on your board.

**S:**  $(\frac{1}{8} + \frac{1}{8}) + (\frac{1}{8} + \frac{1}{8}) + (\frac{1}{8} + \frac{1}{8}) = \frac{6}{8}$ .

**T:** Good. Now, how many eighths is that in all?

**S:** 6 eighths.

**T:** So we can write a multiplication sentence too. How many copies of  $\frac{1}{8}$ ?

**S:** 6 copies.  $6 \times \frac{1}{8} = \frac{6}{8}$ .

**T:** And that tells us  $\frac{3}{4}$  equals  $\frac{6}{8}$ .

**2 Draw an area model to show that two-thirds is equivalent to eight-twelfths, and express the equivalence using addition and multiplication of unit fractions.**

Draw an area model to show that  $\frac{2}{3} = \frac{8}{12}$ . Write an addition sentence using parentheses and a multiplication sentence to prove the equivalence.

*Answer:  $(\frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12}) + (\frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12}) = \frac{8}{12}$ ;  $8 \times \frac{1}{12} = \frac{8}{12}$ ;  $\frac{2}{3} = \frac{8}{12}$*

**T:** This time, you're going to draw the area model. We want to show that  $\frac{2}{3} = \frac{8}{12}$ . Which fraction should you draw first?

**S:**  $\frac{2}{3}$ , because thirds are the bigger pieces.

**T:** Go ahead. Draw a rectangle for 1 whole, divide it into thirds, and shade 2 of them.

**S:** (Draw and shade.)

**T:** Now, we need twelfths. If there are already 3 columns for thirds, how many rows do you need to make 12 equal parts?

**S:** 4 rows. Because 3 times 4 is 12.

**T:** So how many horizontal lines do you need to draw?

**S:** 3 horizontal lines to make 4 rows.

**T:** Draw them across the whole rectangle.

**S:** (Draw.)

**T:** How many total parts are in the whole now?

**S:** 12.

**T:** And how many are shaded?

**S:** 8.

**T:** So  $\frac{2}{3}$  equals  $\frac{8}{12}$ . Now write the addition sentence. Use parentheses to group the twelfths inside each third.

**S:**  $(\frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12}) + (\frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12}) = \frac{8}{12}$ .

**T:** And the multiplication sentence?

**S:**  $8 \times \frac{1}{12} = \frac{8}{12}$ .

**T:** What do those sentences tell you about  $\frac{2}{3}$  and  $\frac{8}{12}$ ?

**S:** They're equivalent. The shaded area is the same amount — just described with smaller pieces.

**3 Draw an area model to represent a fraction greater than one, then partition it into smaller units to find an equivalent fraction.**

Draw an area model to represent  $\frac{5}{3}$ . Then, partition the thirds into sixths to find an equivalent fraction.

*Answer:  $\frac{5}{3} = \frac{10}{6}$*

**T:** Now try this one on your own. Draw an area model for  $\frac{5}{3}$ , then partition the thirds into sixths to find an equivalent fraction.

**S:** (Work.)